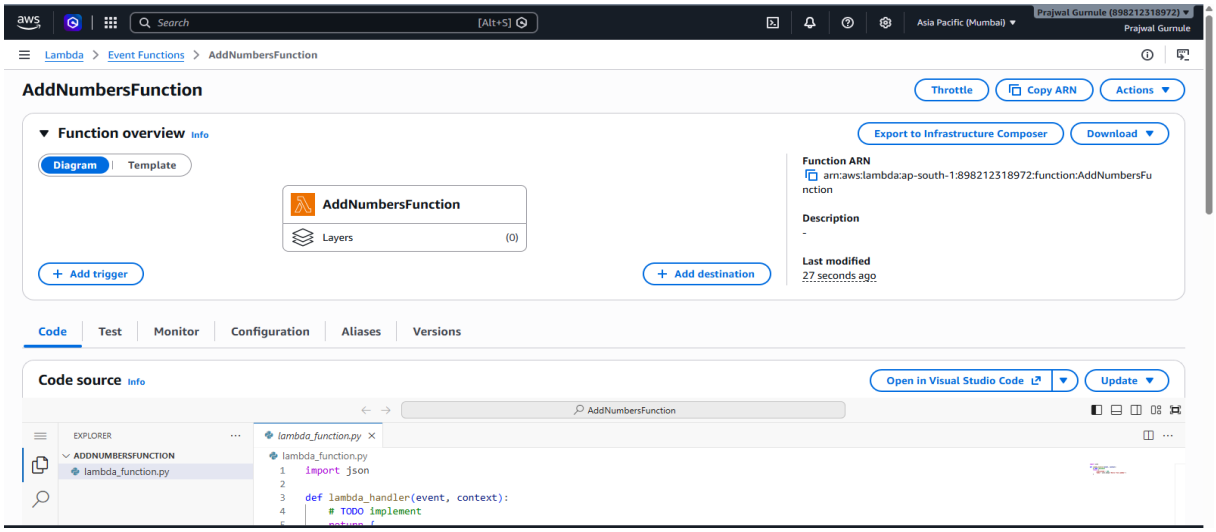
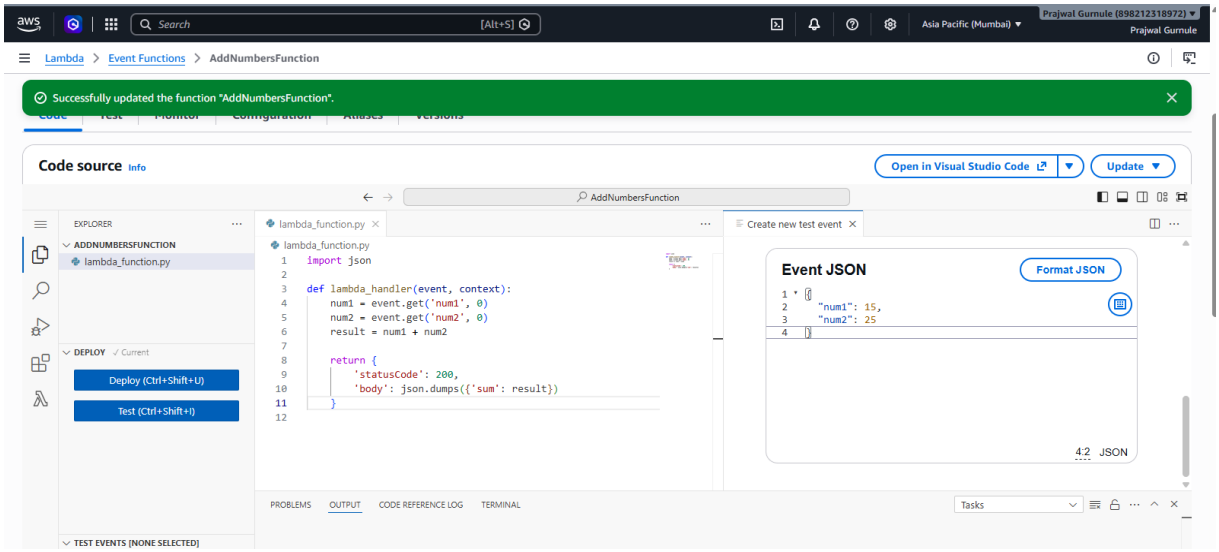
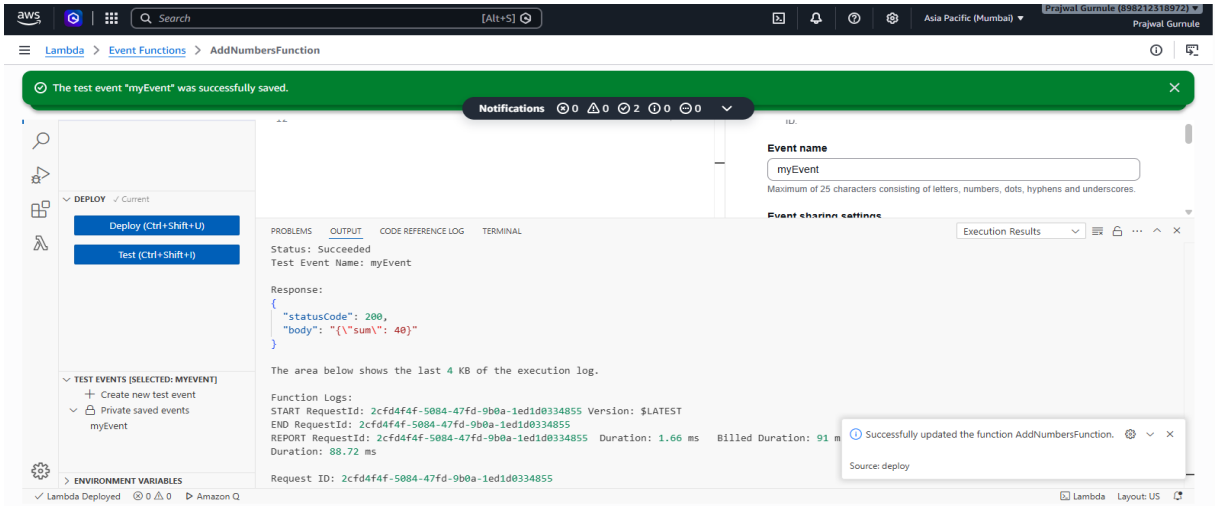
Assignment 9

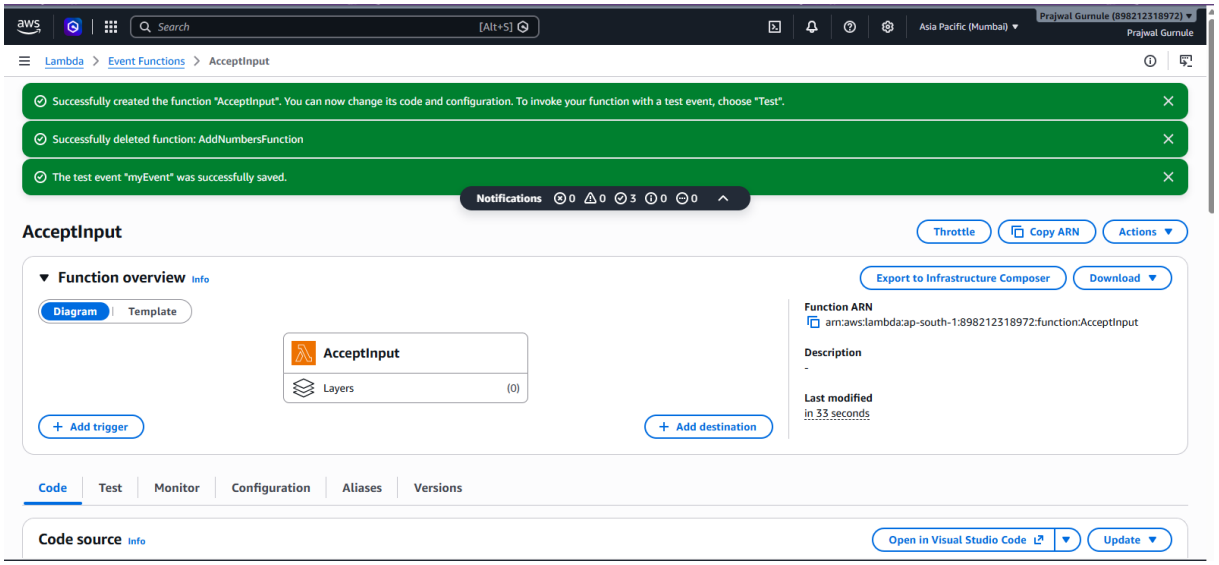
1. Create a AWS Lambda function that takes two numbers as input using a test event and returns their addition.

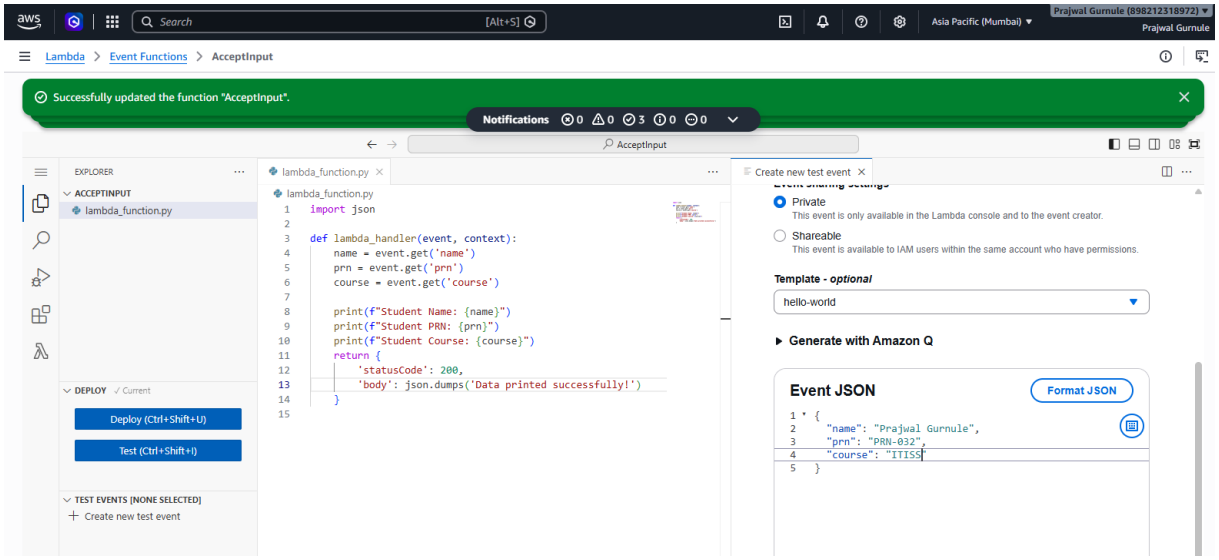


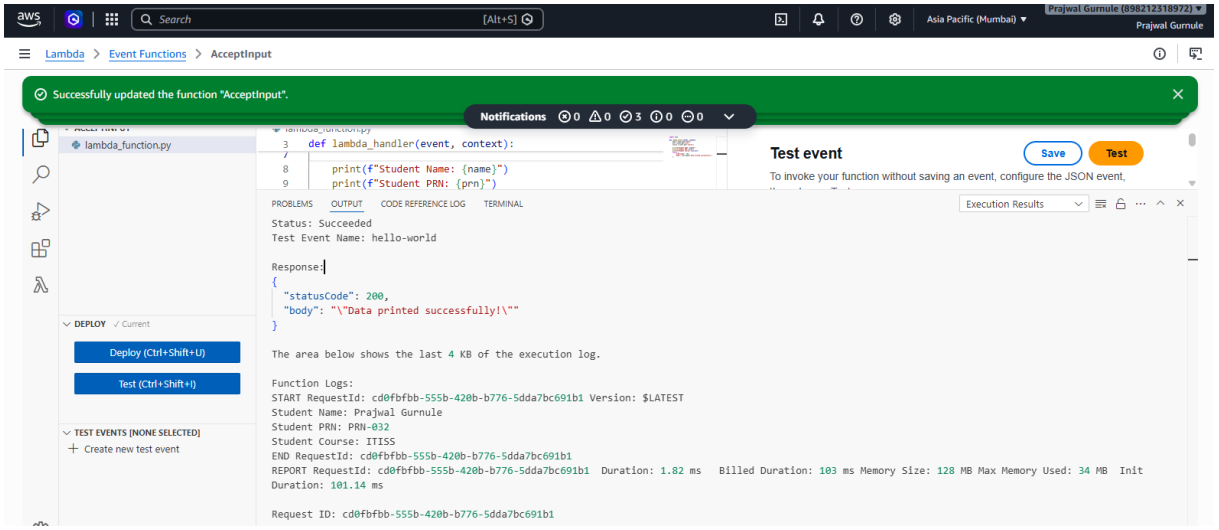


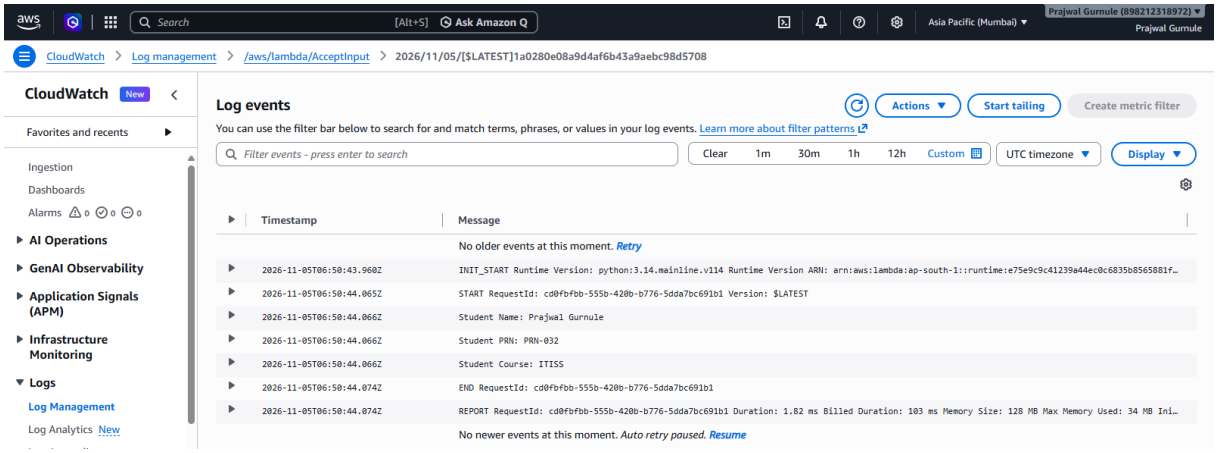


2. Create a Lambda function that accepts input values (name, PRN, course) from a test event and prints them.

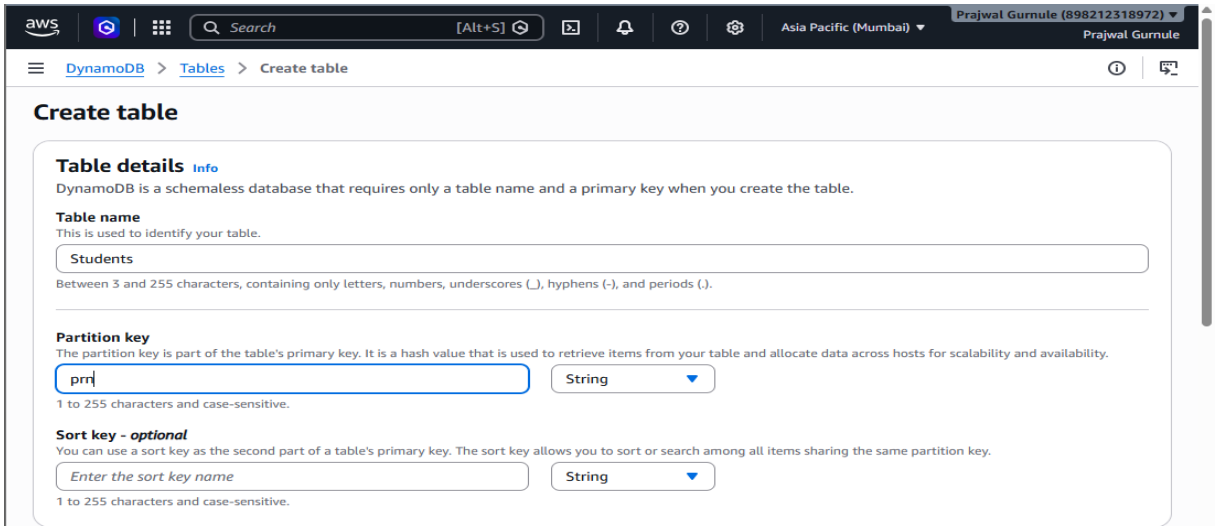


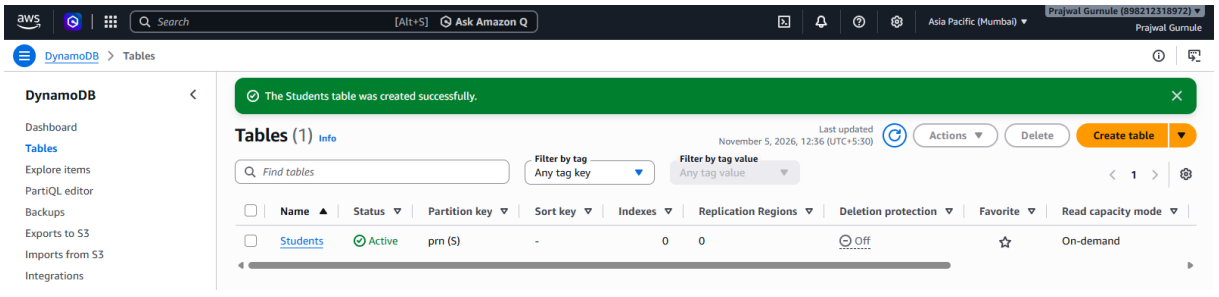


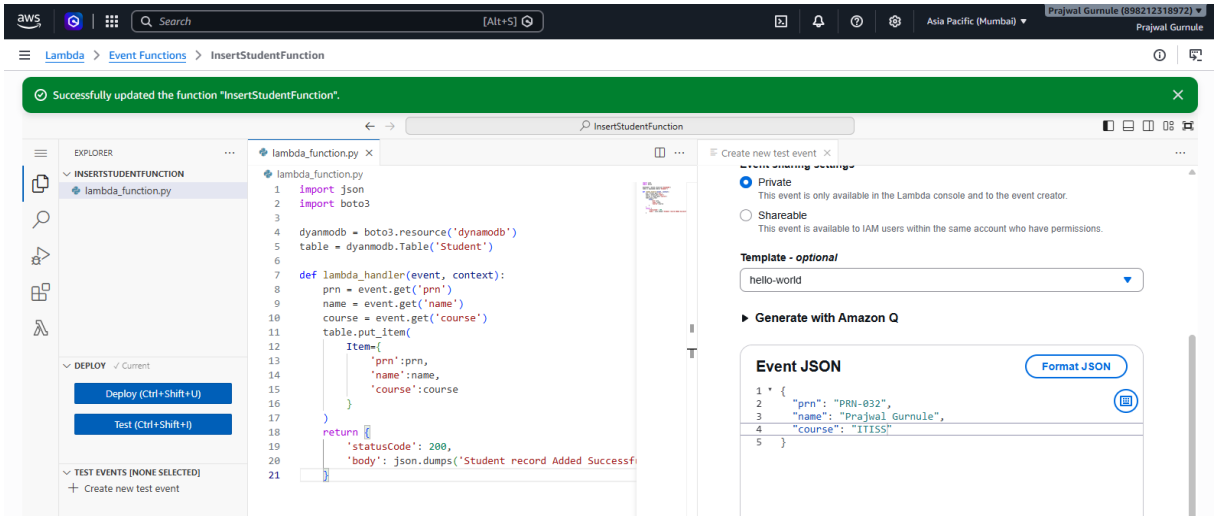


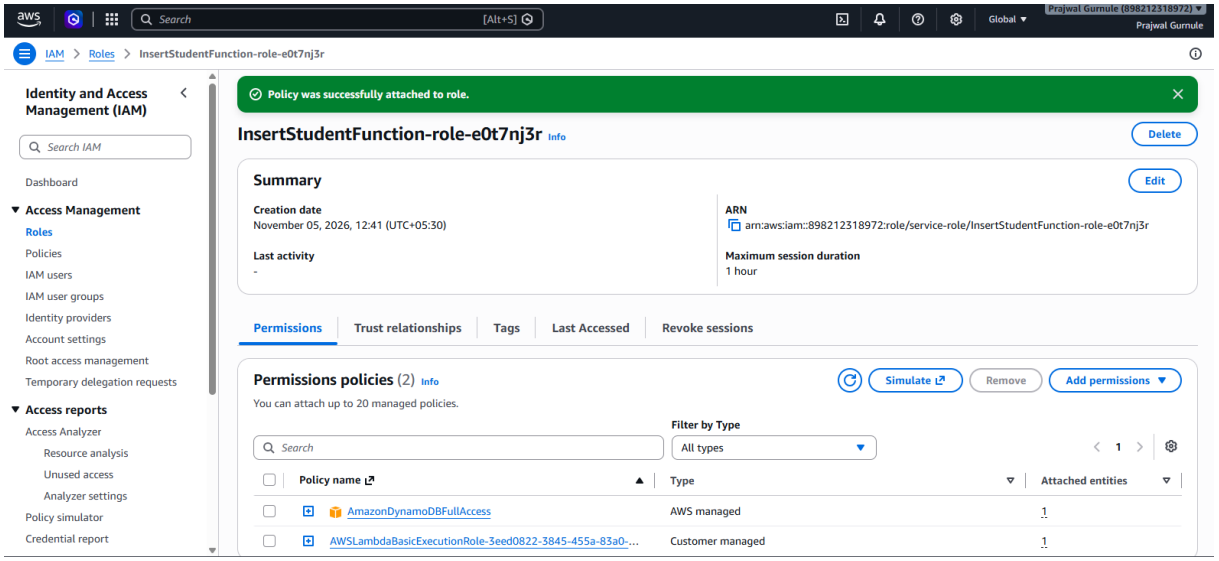


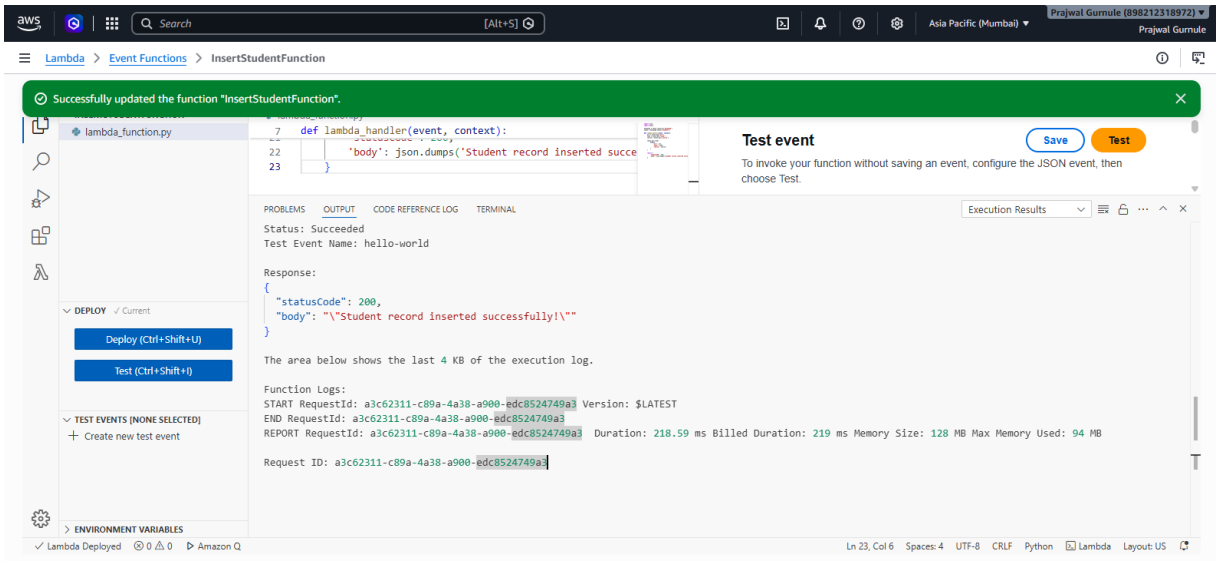
3. Create a Lambda function that inserts student details into a DynamoDB table.

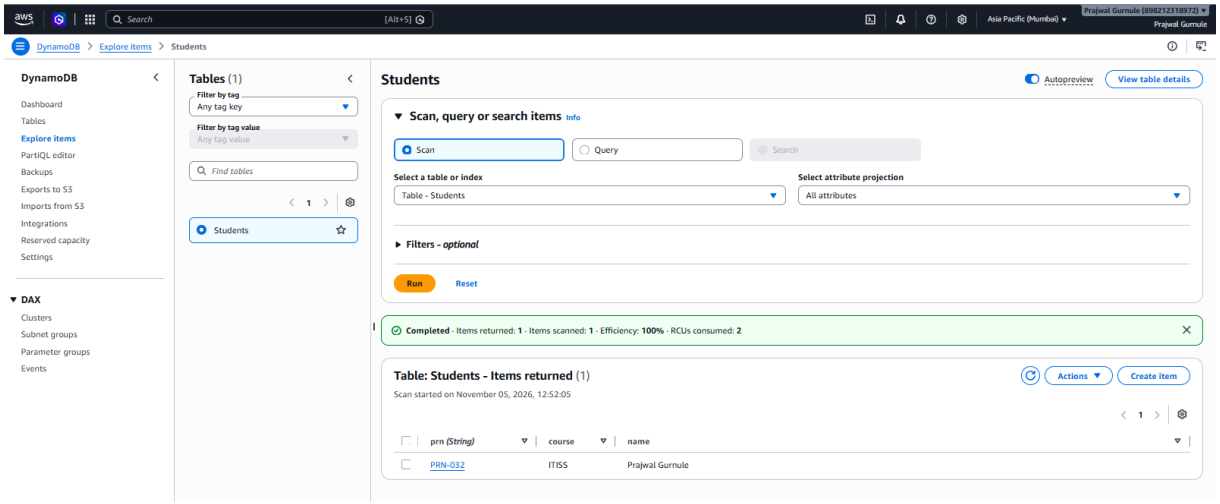




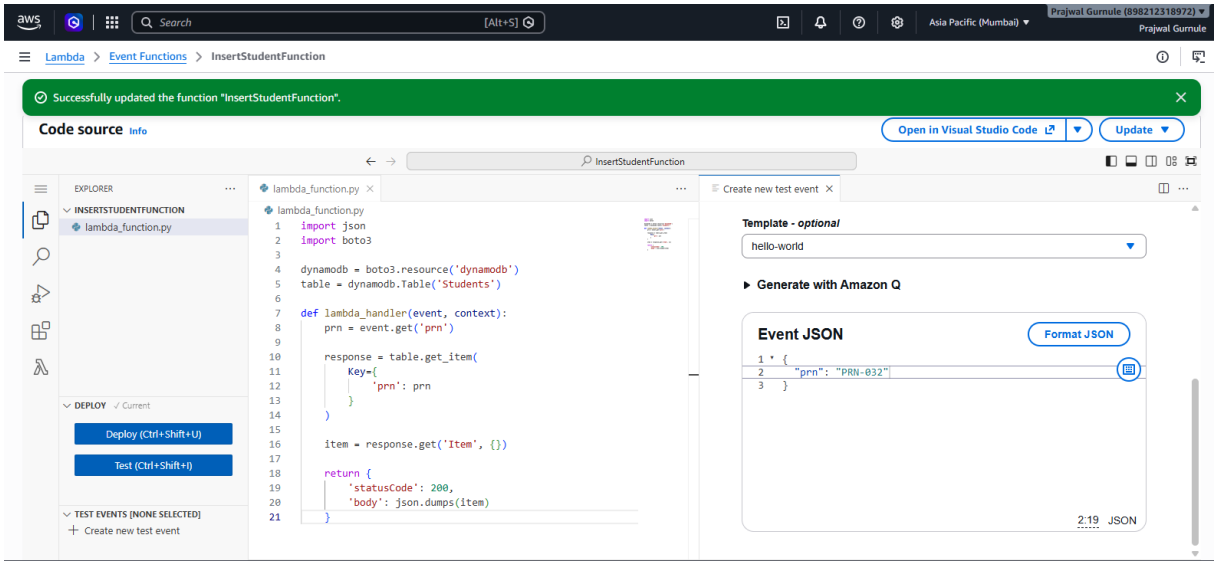


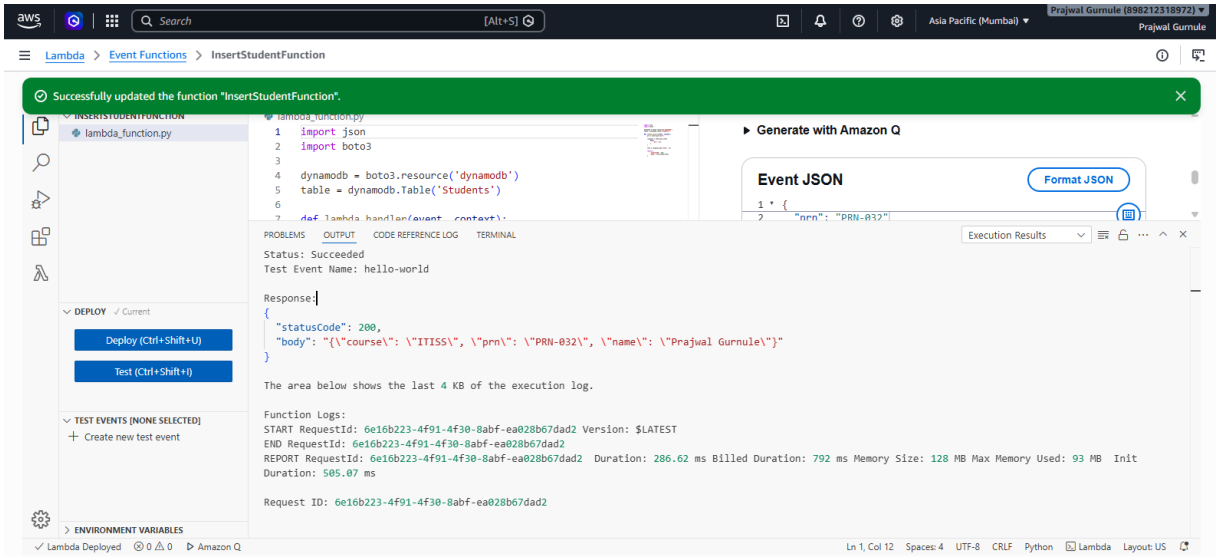

IAM > Roles > InsertStudentFunction-role-e0t7nj3r
Policy was successfully attached to role.
InsertStudentFunction-role-e0t7nj3r Info
Delete
Summary
Edit
Creation date
November 05, 2026, 12:41 (UTC+05:30)
ARN
arn:aws:iam::898212318972:role/service-role/InsertStudentFunction-role-e0t7nj3r
Last activity
-
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (2) Info
You can attach up to 20 managed policies.
Simulate
Remove
Add permissions
Filter by Type
All types
Policy name
Type
Attached entities
AmazonDynamoDBFullAccess
AWS managed
AWSLambdaBasicExecutionRole-3eed0822-3845-455a-83a0-...
Customer managed


Lambda > Event Functions > InsertStudentFunction
Successfully updated the function "InsertStudentFunction".
lambda_function.py
Test event
Save
Test
To invoke your function without saving an event, configure the JSON event, then choose Test.
PROBLEMS
OUTPUT
CODE REFERENCE LOG
TERMINAL
Execution Results
Status: Succeeded
Test Event Name: hello-world
Response:
{
"statusCode": 200,
"body": "\"Student record inserted successfully!\""
}
The area below shows the last 4 KB of the execution log.
Function Logs:
START RequestId: a3c62311-c89a-4a38-a900-edc8524749a3 Version: $LATEST
END RequestId: a3c62311-c89a-4a38-a900-edc8524749a3
REPORT RequestId: a3c62311-c89a-4a38-a900-edc8524749a3 Duration: 218.59 ms Billed Duration: 219 ms Memory Size: 128 MB Max Memory Used: 94 MB
Request ID: a3c62311-c89a-4a38-a900-edc8524749a3
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [NONE SELECTED]
Create new test event
ENVIRONMENT VARIABLES

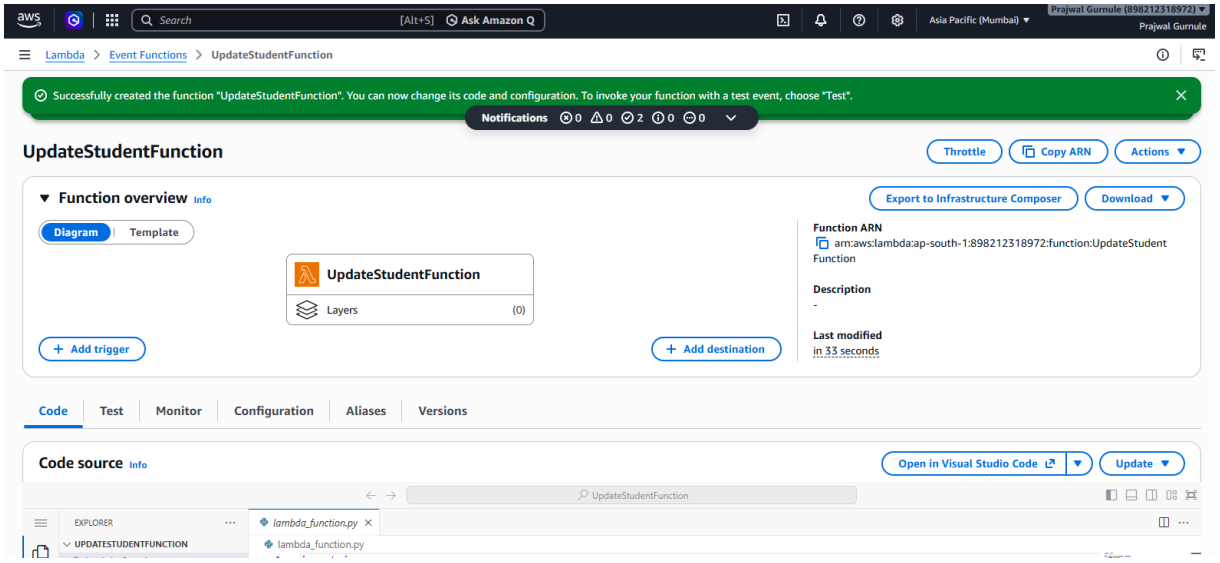

DynamoDB > Explore items > Students
DynamoDB
Tables (1)
Students
Autopreview
View table details
Scan, query or search items Info
Scan
Query
Select a table or index
Table - Students
Select attribute projection
All attributes
Filters - optional
Run
Reset
Completed - Items returned: 1 - Items scanned: 1 - Efficiency: 100% - RCUs consumed: 2
Table: Students - Items returned (1)
Scan started on November 05, 2026, 12:52:05
Actions
Create item
prn (String)
course
name
PRN-032
ITISS
Prajwal Gurnule

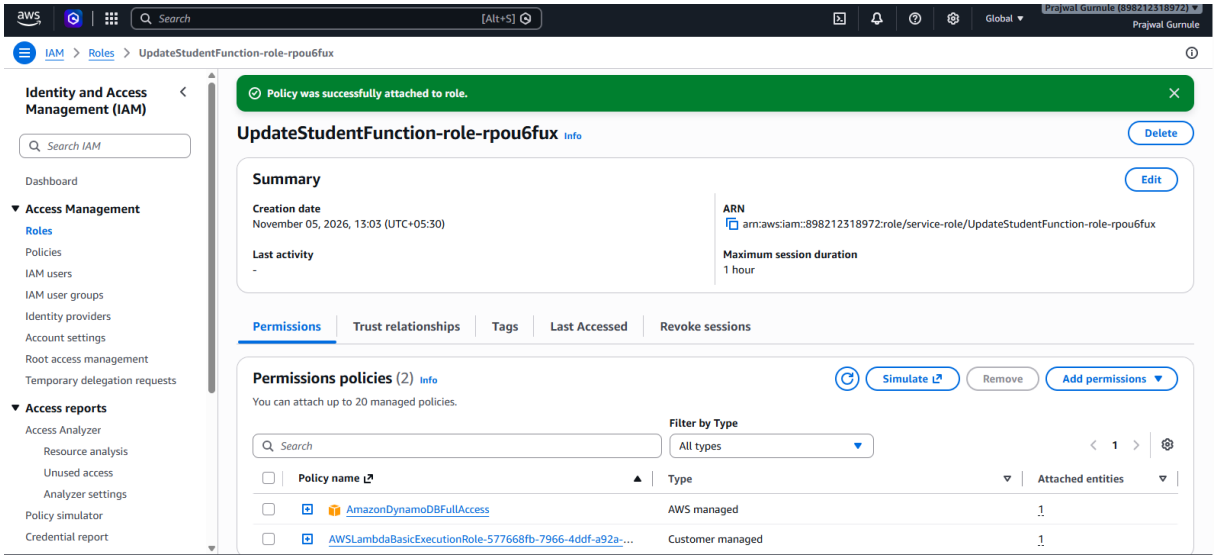
4. Create a Lambda function that reads student details from DynamoDB using PRN.

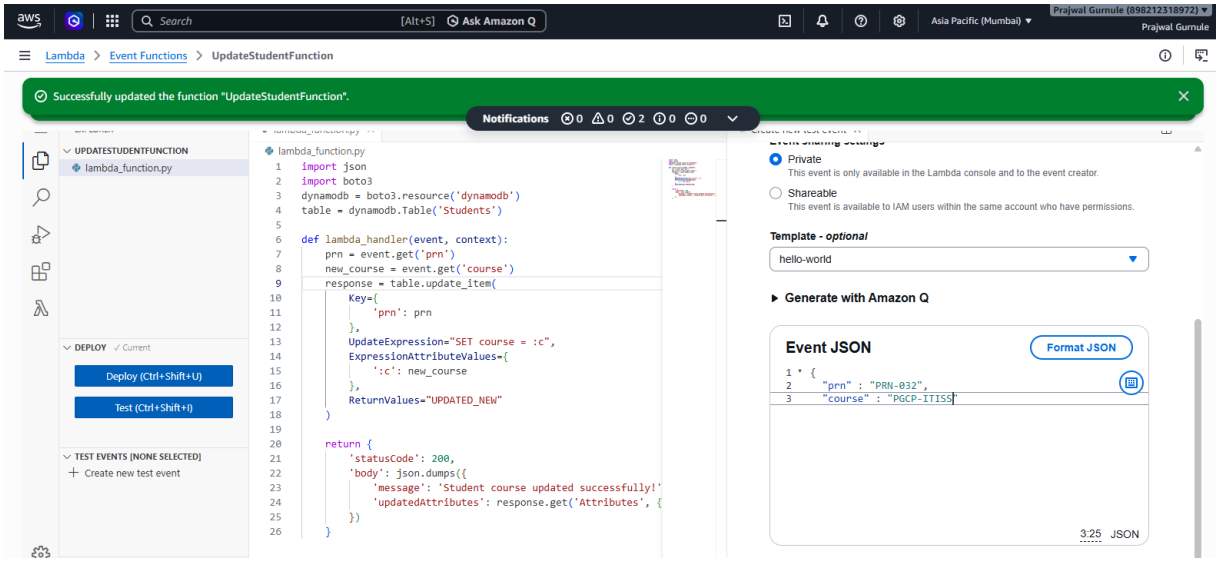


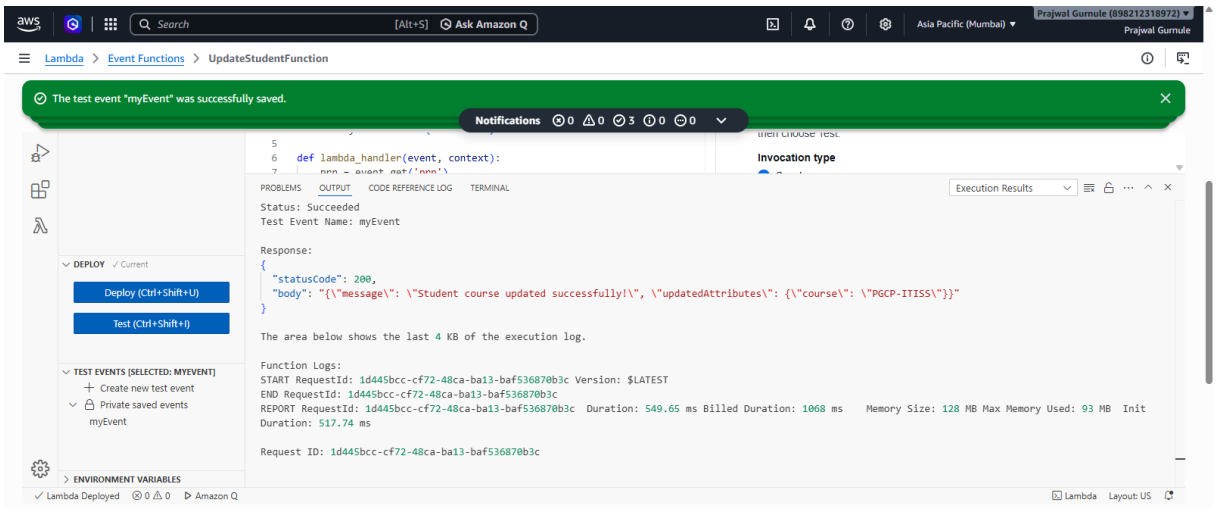


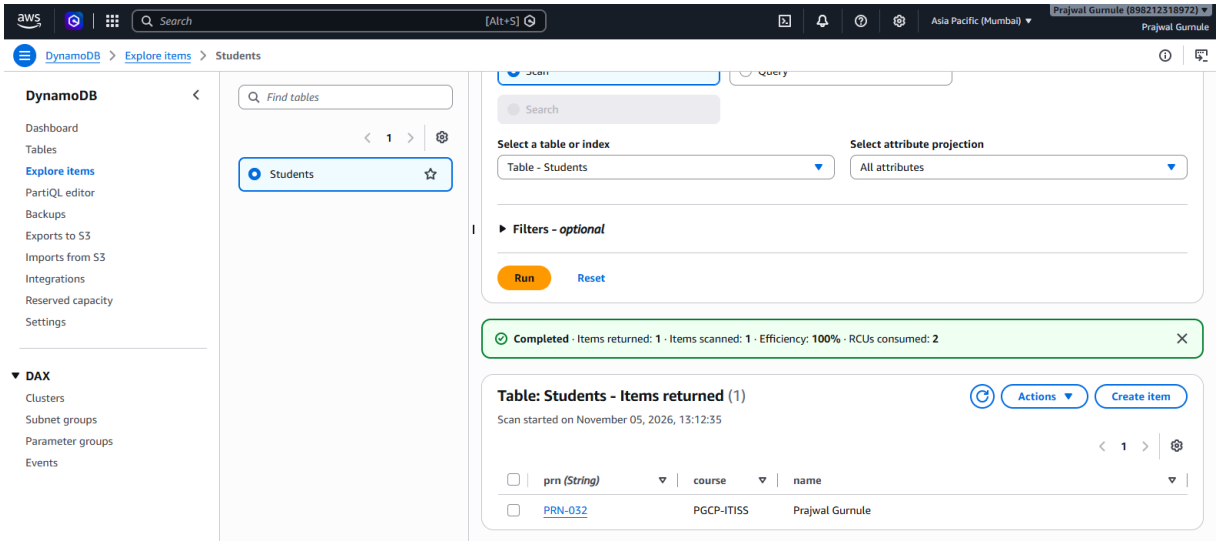
5. Create a Lambda function that updates a student's course name in DynamoDB.



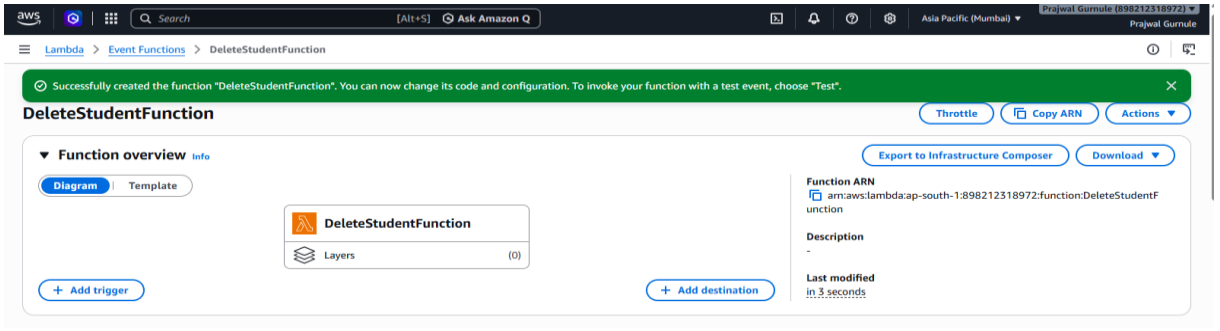

Policy was successfully attached to role.
UpdateStudentFunction-role-rpou6fux
Summary
Creation date
November 05, 2026, 13:03 (UTC+05:30)
ARN
arn:aws:iam::898212318972:role/service-role/UpdateStudentFunction-role-rpou6fux
Last activity
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (2)
You can attach up to 20 managed policies.
Simulate
Remove
Add permissions
Filter by Type
All types
Policy name
Type
Attached entities
AmazonDynamoDBFullAccess
AWS managed
AWSLambdaBasicExecutionRole-577668fb-7966-4ddf-a92a-...
Customer managed

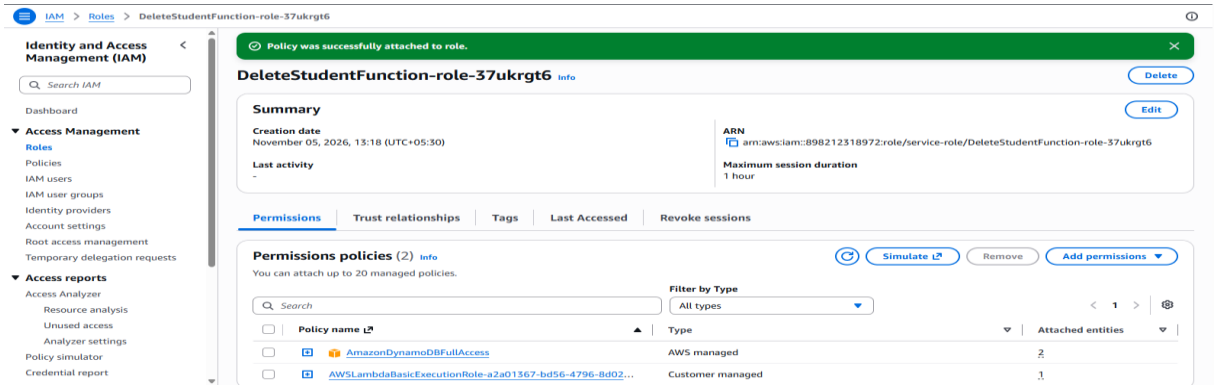

Successfully updated the function "UpdateStudentFunction".
Notifications
lambda_function.py
import json
import boto3
dynamodb = boto3.resource('dynamodb')
table = dynamodb.Table('Students')
def lambda_handler(event, context):
prn = event.get('prn')
new_course = event.get('course')
response = table.update_item(
Key={
'prn': prn
},
UpdateExpression="SET course = :c",
ExpressionAttributeValues={
':c': new_course
},
ReturnValues="UPDATED_NEW"
)
return {
'statusCode': 200,
'body': json.dumps({
'message': 'Student course updated successfully!'
'updatedAttributes': response.get('Attributes', {
})
}
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Private
This event is only available in the Lambda console and to the event creator.
Shareable
This event is available to IAM users within the same account who have permissions.
Template - optional
hello-world
Generate with Amazon Q
Event JSON
Format JSON
{
"prn" : "PRN-032",
"course" : "PGCP-ITISS"
}

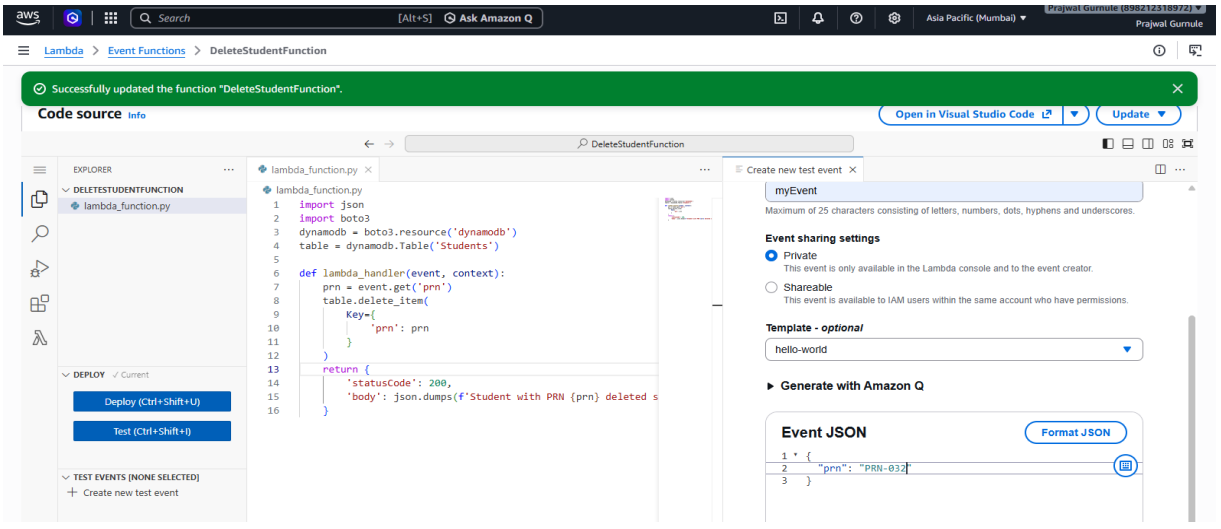

The test event "myEvent" was successfully saved.
Notifications
Invocation type
PROBLEMS
OUTPUT
CODE REFERENCE LOG
TERMINAL
Execution Results
Status: Succeeded
Test Event Name: myEvent
Response:
{
"statusCode": 200,
"body": "{\"message\": \"Student course updated successfully!\", \"updatedAttributes\": {\"course\": \"PGCP-ITISS\"}}"
}
The area below shows the last 4 KB of the execution log.
Function Logs:
START RequestId: 1d445bcc-cf72-48ca-ba13-baf536870b3c Version: $LATEST
END RequestId: 1d445bcc-cf72-48ca-ba13-baf536870b3c
REPORT RequestId: 1d445bcc-cf72-48ca-ba13-baf536870b3c Duration: 549.65 ms Billed Duration: 1068 ms Memory Size: 128 MB Max Memory Used: 93 MB Init Duration: 517.74 ms
Request ID: 1d445bcc-cf72-48ca-ba13-baf536870b3c
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Create new test event
Private saved events
myEvent

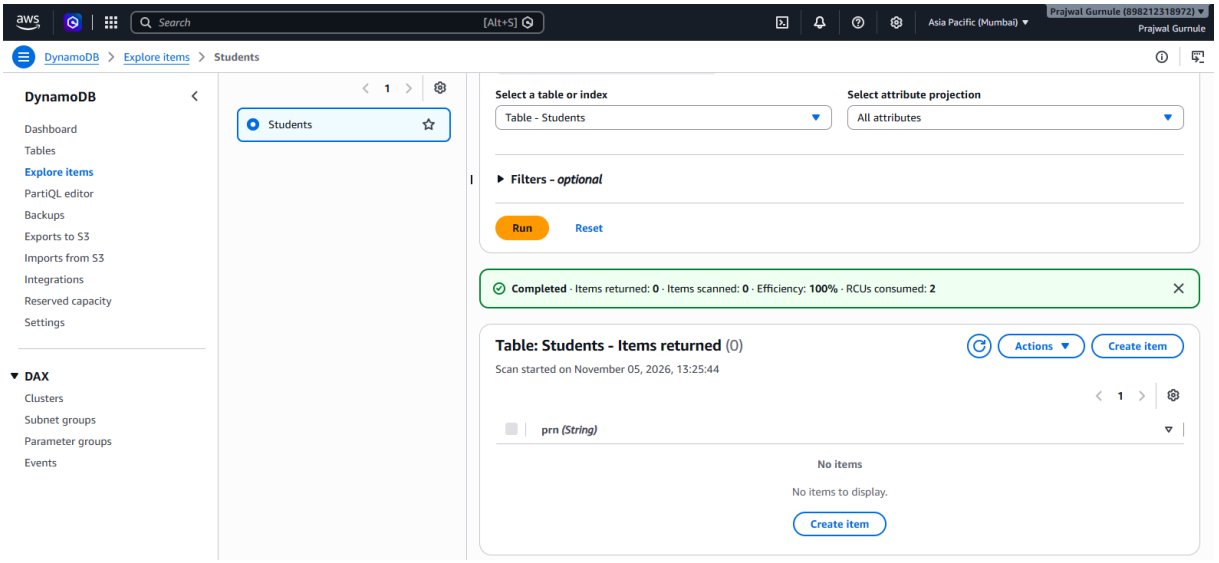


6. Create a Lambda function that deletes a student record from DynamoDB using PRN.

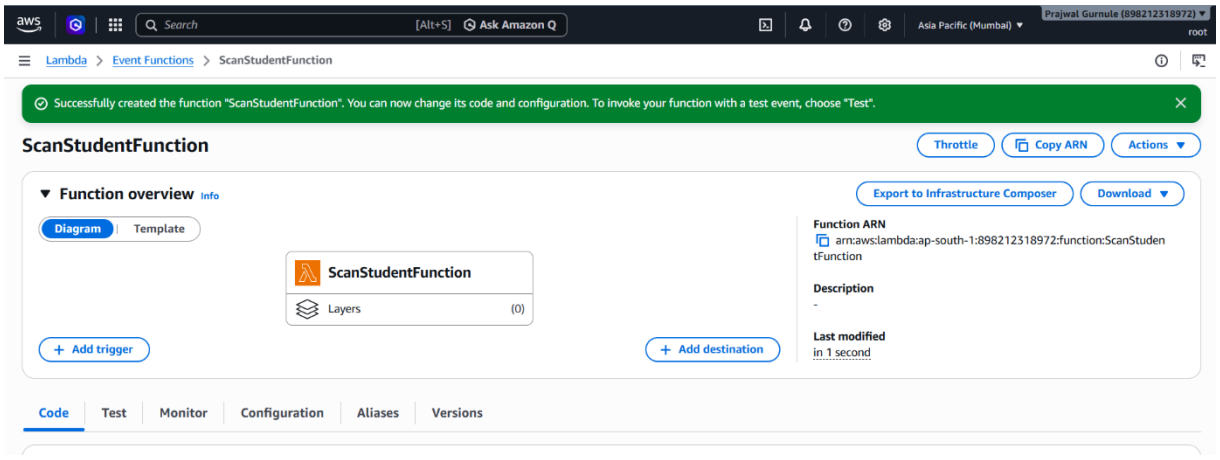


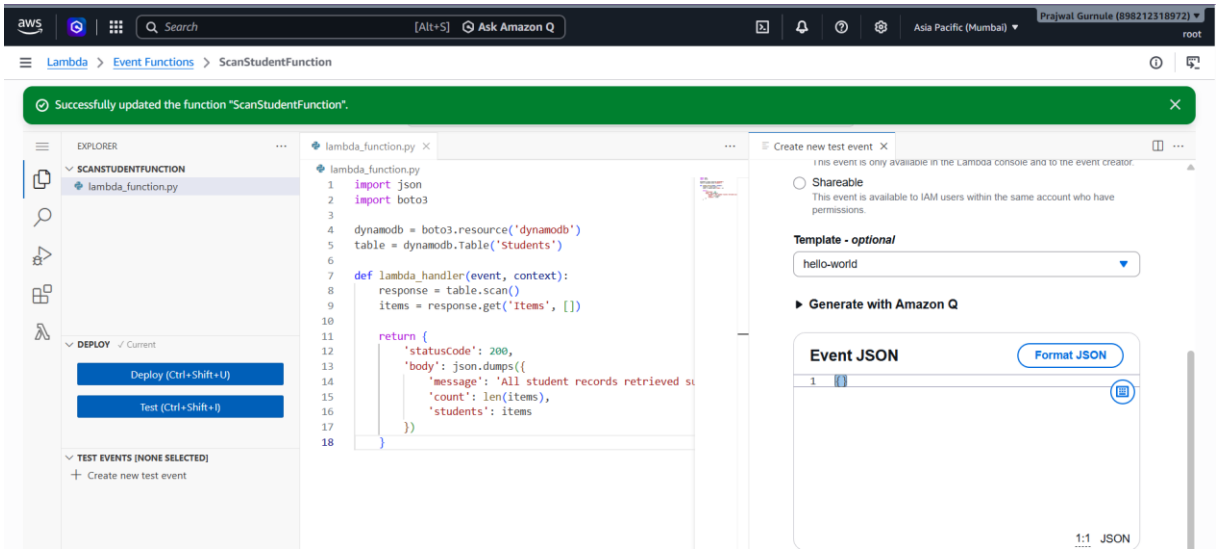


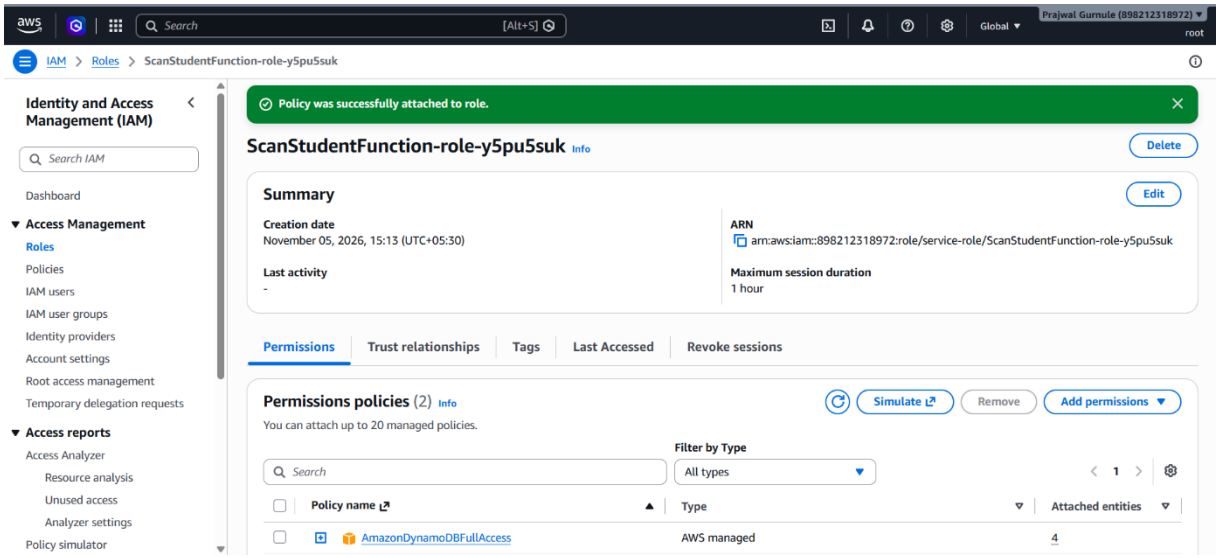


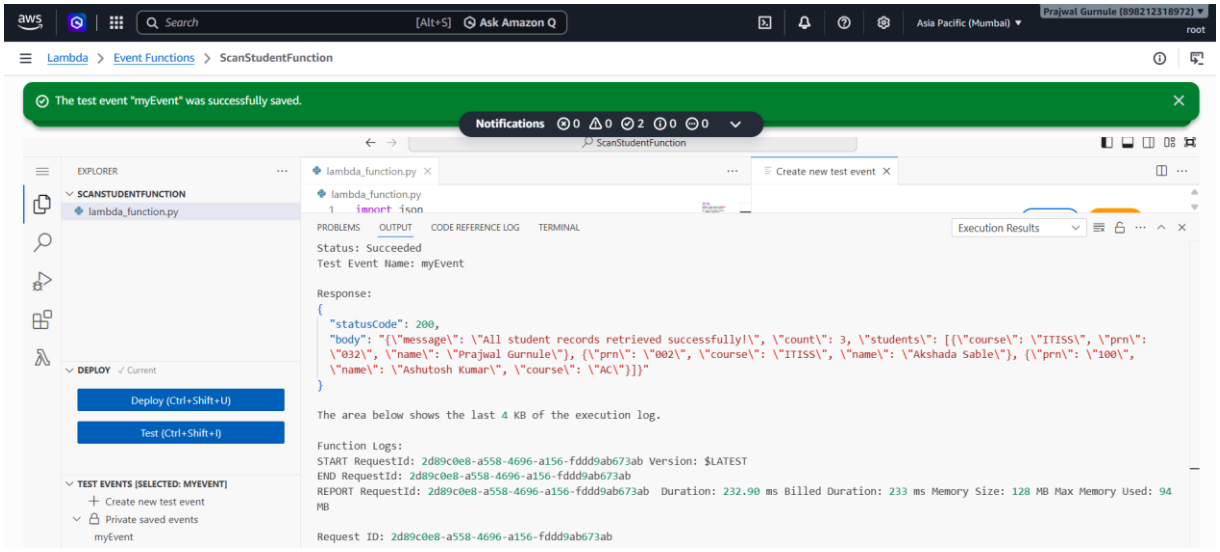


7. Create a Lambda function that scans all student records from a DynamoDB table.

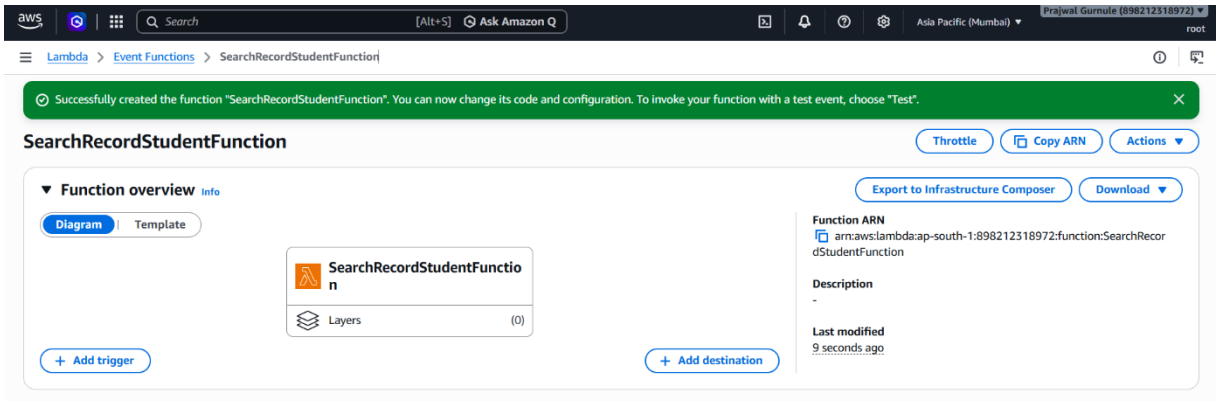


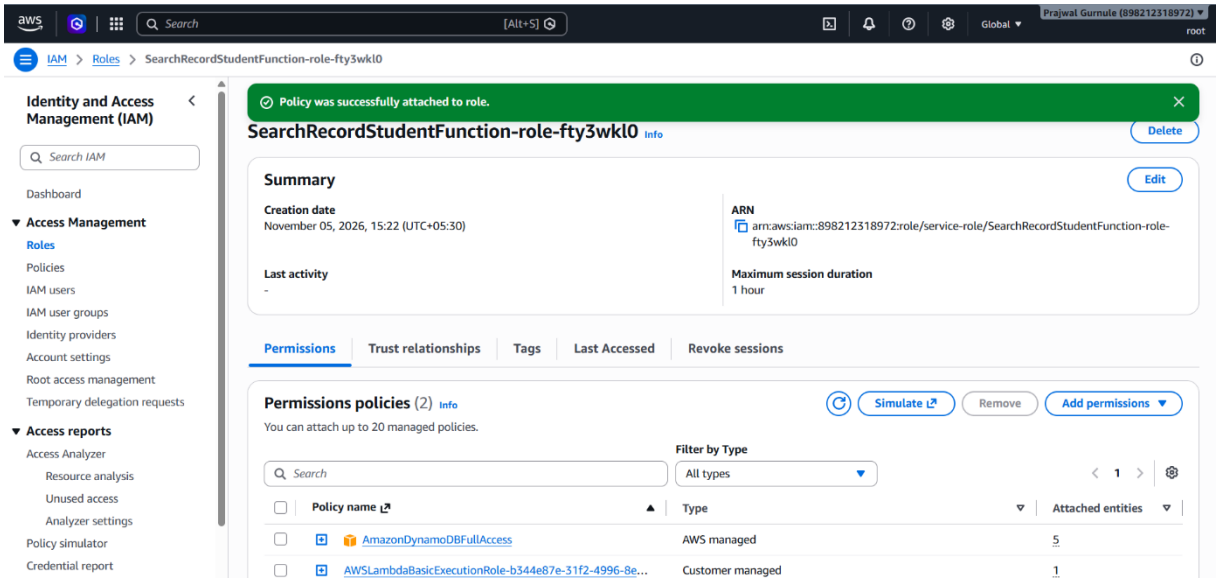


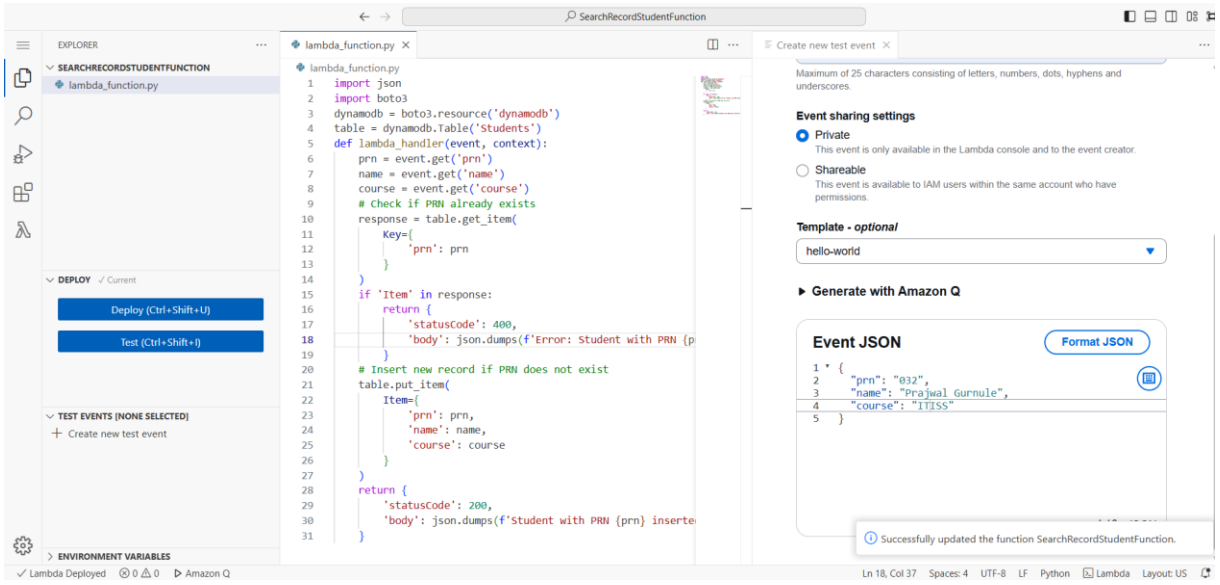


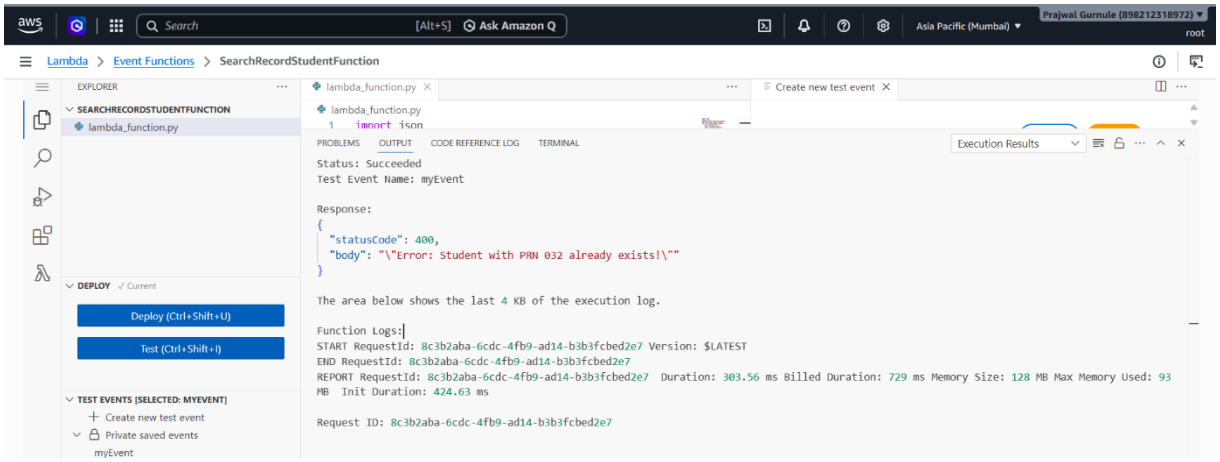


8. Create a Lambda function that checks whether a student PRN already exists before inserting a new record.

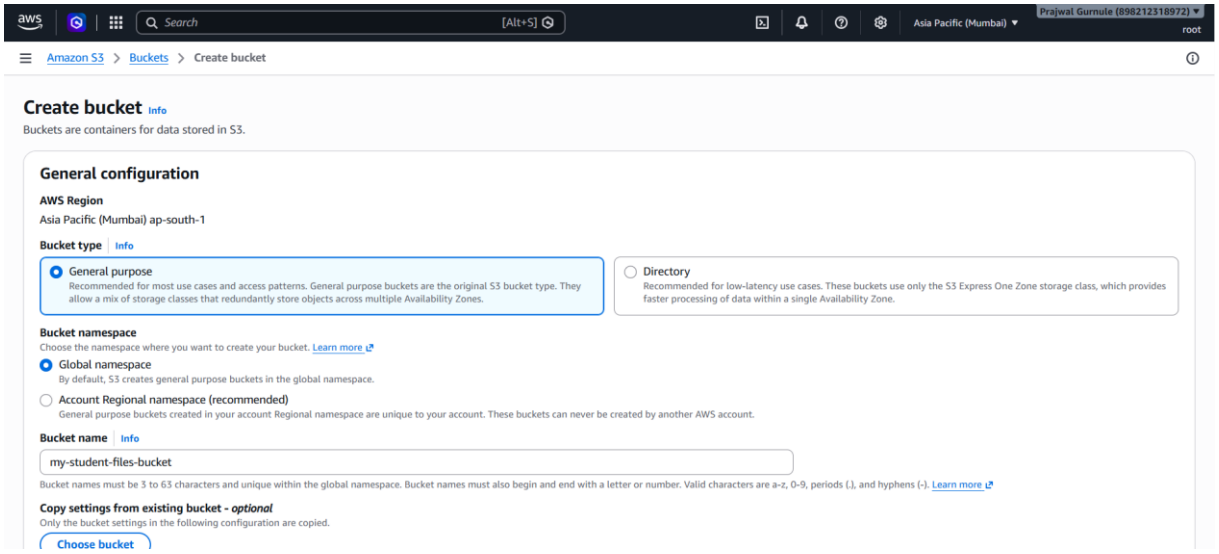


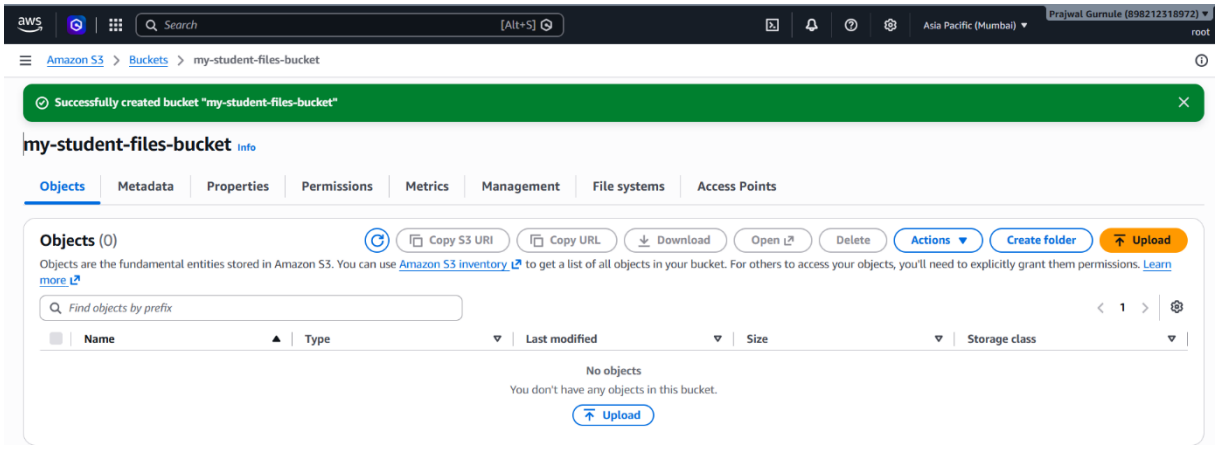

Policy was successfully attached to role.
SearchRecordStudentFunction-role-fty3wkl0
Delete
Summary
Edit
Creation date
November 05, 2026, 15:22 (UTC+05:30)
ARN
arn:aws:iam::898212318972:role/service-role/SearchRecordStudentFunction-role-fty3wkl0
Last activity
-
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (2)
You can attach up to 20 managed policies.
Simulate
Remove
Add permissions
Filter by Type
All types
Policy name
Type
Attached entities
AmazonDynamoDBFullAccess
AWS managed
5
AWSLambdaBasicExecutionRole-b344e87e-31f2-4996-8e...
Customer managed
1

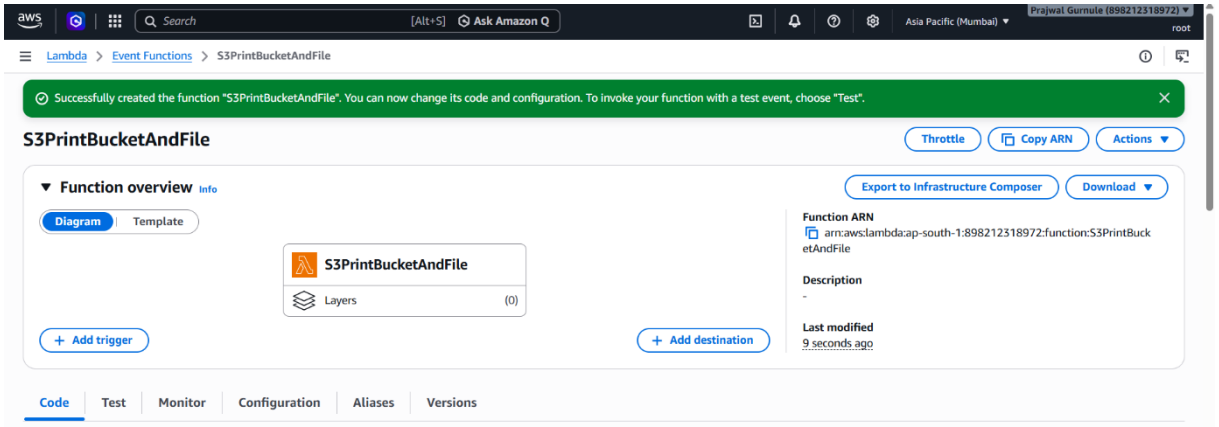

SearchRecordStudentFunction
lambda_function.py
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Create new test event
Event sharing settings
Private
Shareable
Template - optional
hello-world
Generate with Amazon Q
Event JSON
Format JSON
Successfully updated the function SearchRecordStudentFunction.

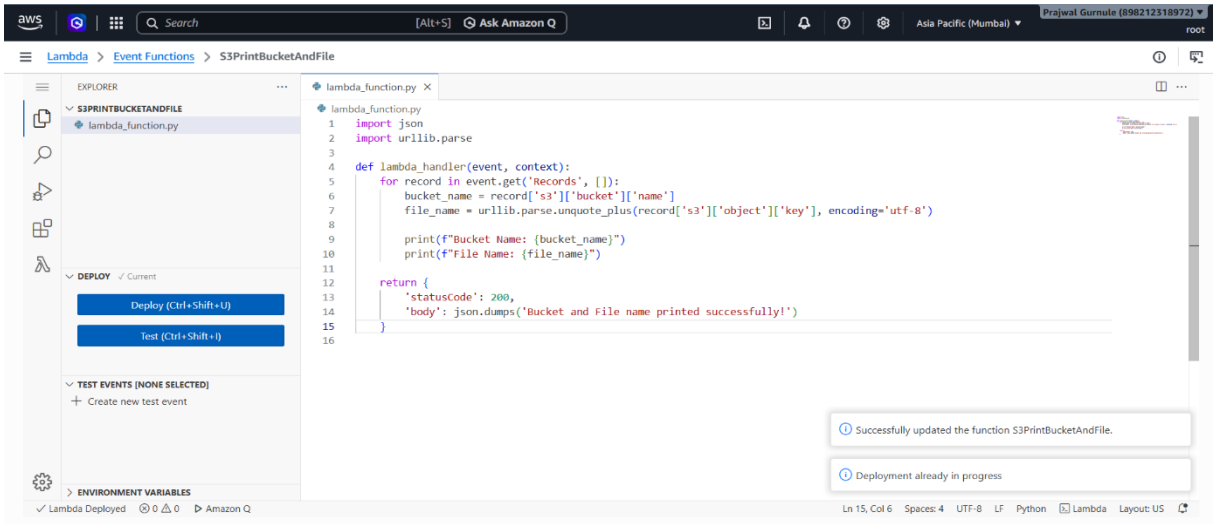

Lambda
Event Functions
SearchRecordStudentFunction
Execution Results
Status: Succeeded
Test Event Name: myEvent
Response:
"statusCode": 400,
"body": "\"Error: Student with PRN 032 already exists!\""
The area below shows the last 4 KB of the execution log.
Function Logs:
START RequestId: 8c3b2aba-6cdc-4fb9-ad14-b3b3fcbed2e7 Version: $LATEST
END RequestId: 8c3b2aba-6cdc-4fb9-ad14-b3b3fcbed2e7
REPORT RequestId: 8c3b2aba-6cdc-4fb9-ad14-b3b3fcbed2e7 Duration: 303.56 ms Billed Duration: 729 ms Memory Size: 128 MB Max Memory Used: 93 MB Init Duration: 424.63 ms
Request ID: 8c3b2aba-6cdc-4fb9-ad14-b3b3fcbed2e7

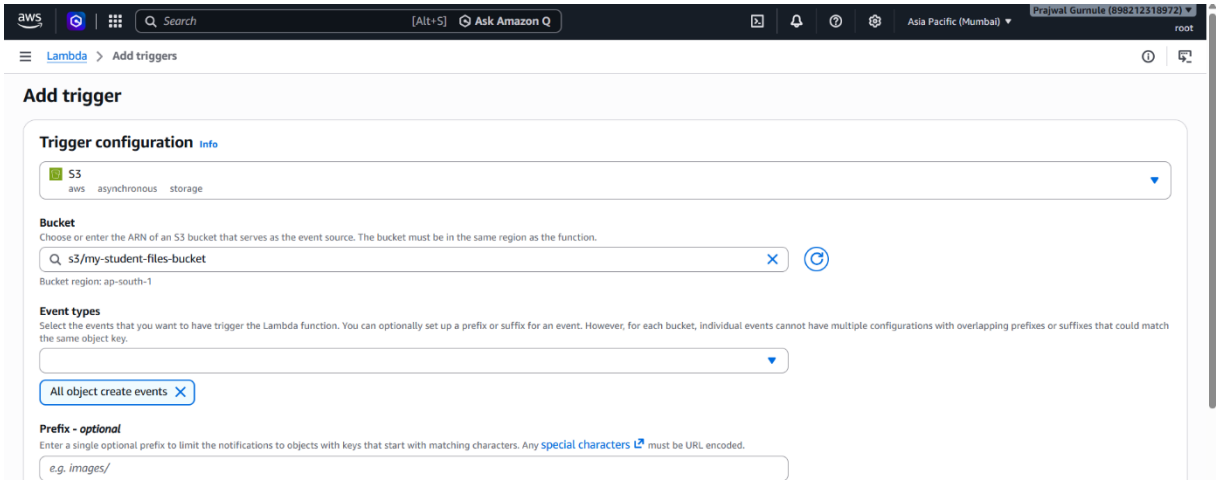
9. Create a Lambda function that triggers when a file is uploaded to an Amazon S3 bucket and prints the bucket name and file name.

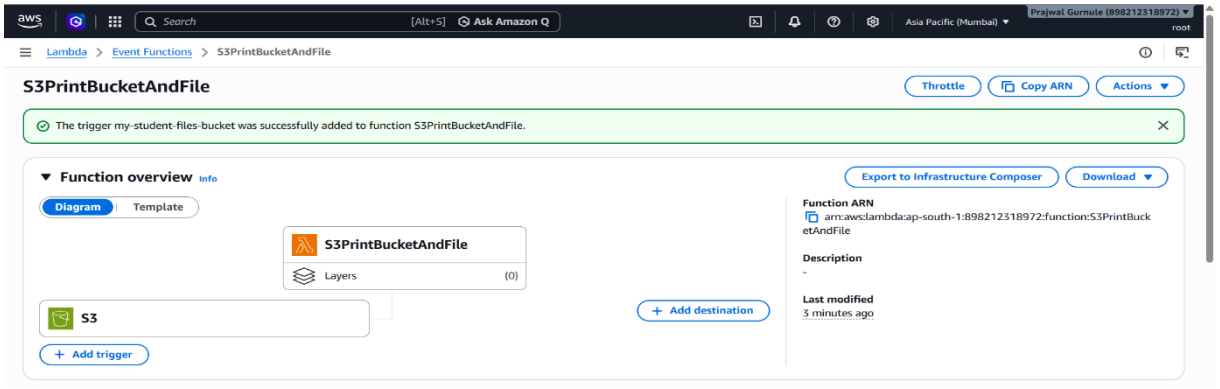


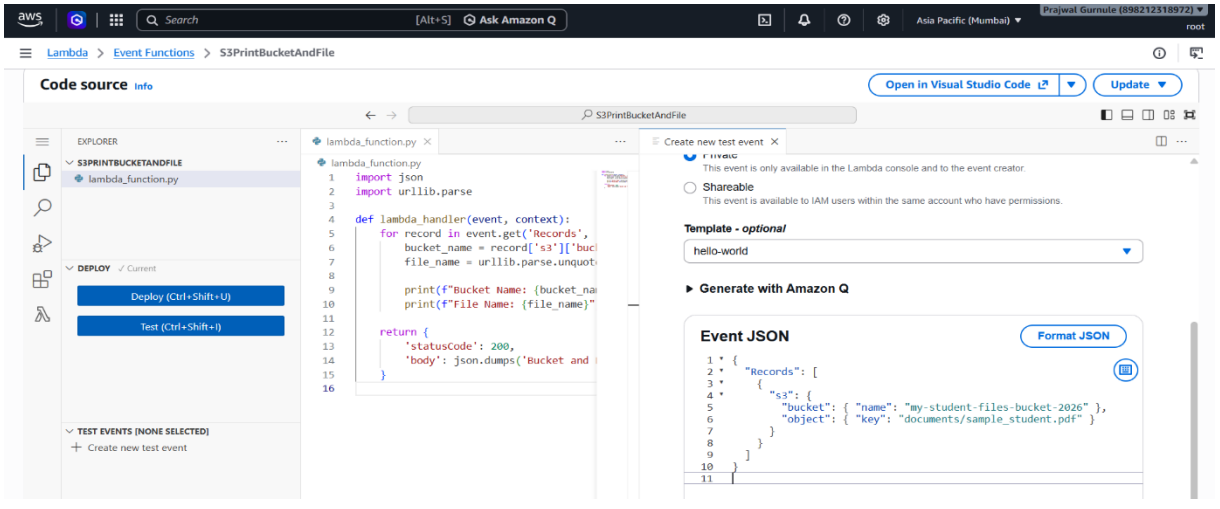


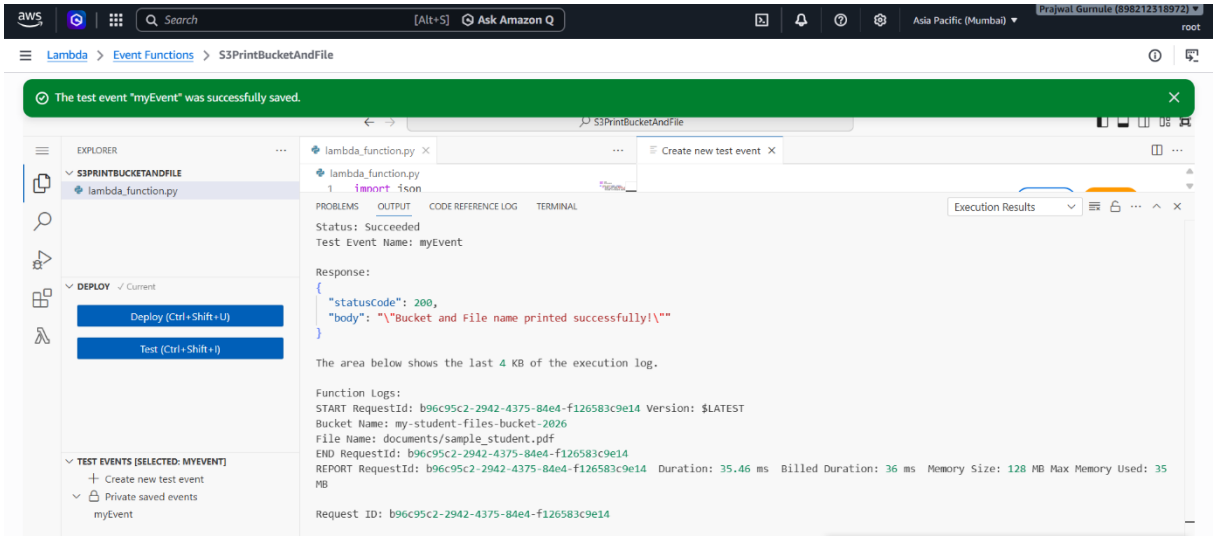


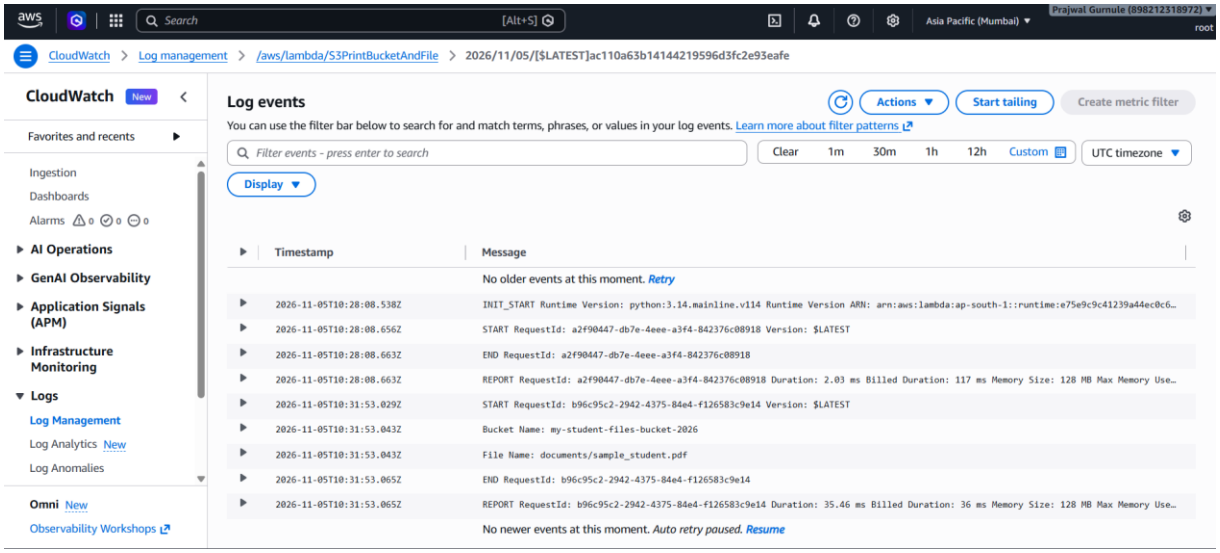

Lambda > Event Functions > S3PrintBucketAndFile
lambda_function.py
import json
import urllib.parse
def lambda_handler(event, context):
for record in event.get('Records', []):
bucket_name = record['s3']['bucket']['name']
file_name = urllib.parse.unquote_plus(record['s3']['object']['key'], encoding='utf-8')
print(f"Bucket Name: {bucket_name}")
print(f"File Name: {file_name}")
return {
'statusCode': 200,
'body': json.dumps('Bucket and File name printed successfully!')
}
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Create new test event
Successfully updated the function S3PrintBucketAndFile.
Deployment already in progress


Lambda > Add triggers
Add trigger
Trigger configuration Info
S3
aws asynchronous storage
Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.
s3/my-student-files-bucket
Bucket region: ap-south-1
Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.
All object create events
Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.
e.g. images/

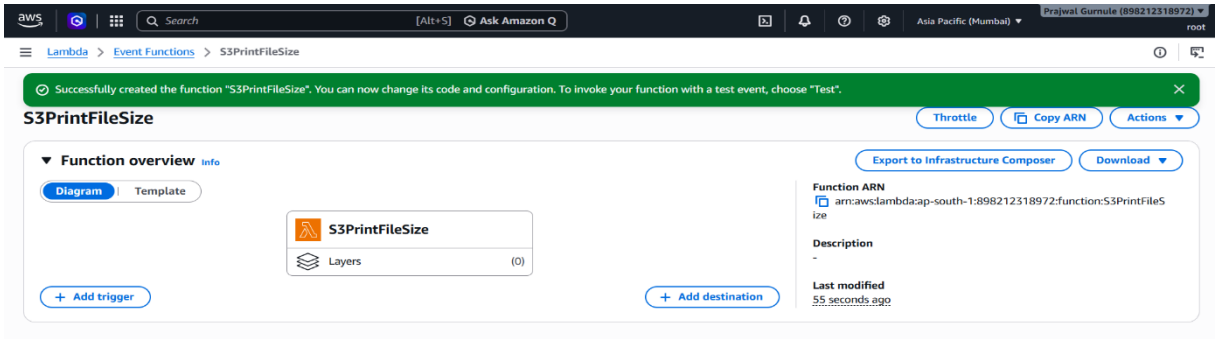

Lambda > Event Functions > S3PrintBucketAndFile
S3PrintBucketAndFile
Throttle
Copy ARN
Actions
The trigger my-student-files-bucket was successfully added to function S3PrintBucketAndFile.
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
S3PrintBucketAndFile
Layers (0)
S3
Add destination
Add trigger
Function ARN
arn:aws:lambda:ap-south-1:898212318972:function:S3PrintBucketAndFile
Description
Last modified
5 minutes ago

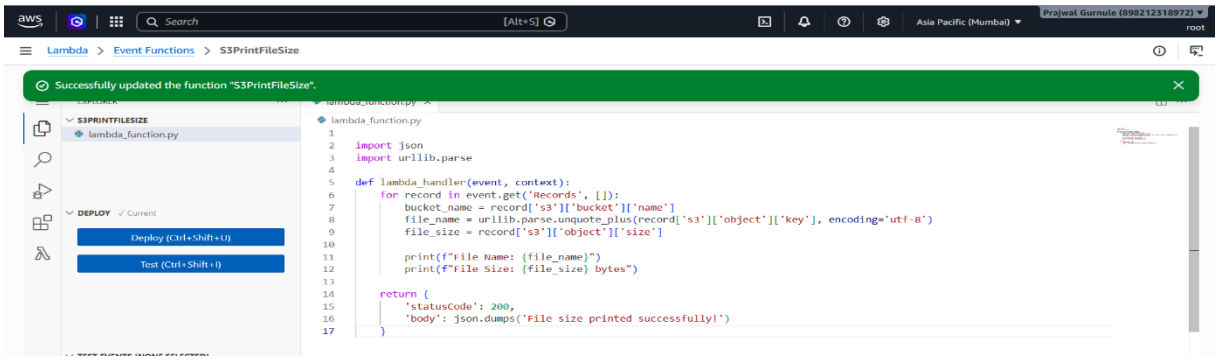

Lambda > Event Functions > S3PrintBucketAndFile
Code source Info
Open in Visual Studio Code
Update
lambda_function.py
Create new test event
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Shareable
This event is available to IAM users within the same account who have permissions.
Template - optional
hello-world
Generate with Amazon Q
Event JSON
Format JSON
{
"Records": [
{
"s3": {
"bucket": { "name": "my-student-files-bucket-2026" },
"object": { "key": "documents/sample_student.pdf" }
}
}
]
}

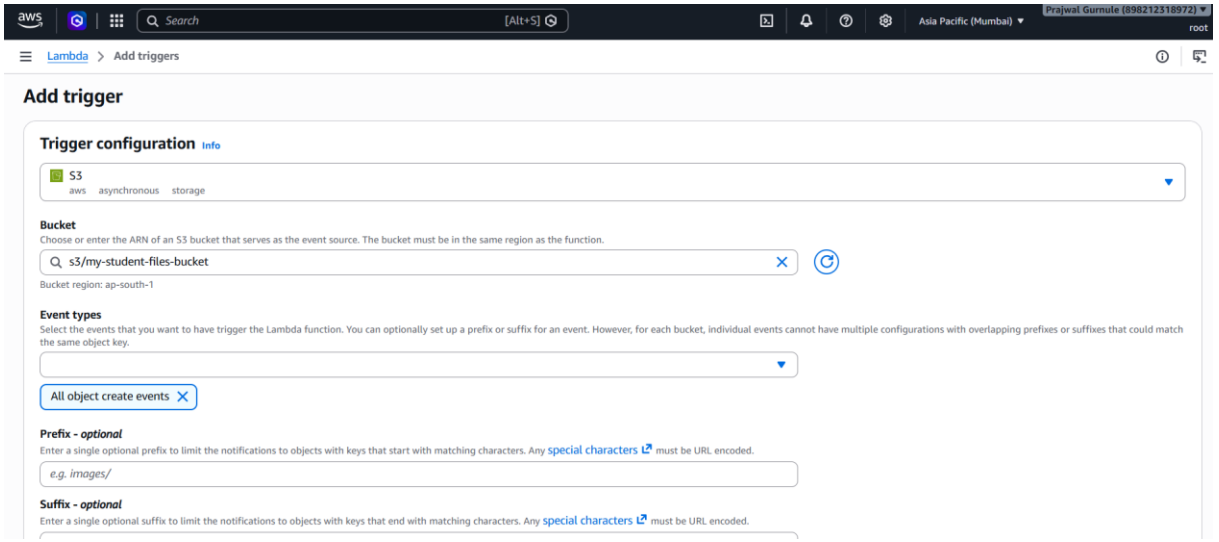


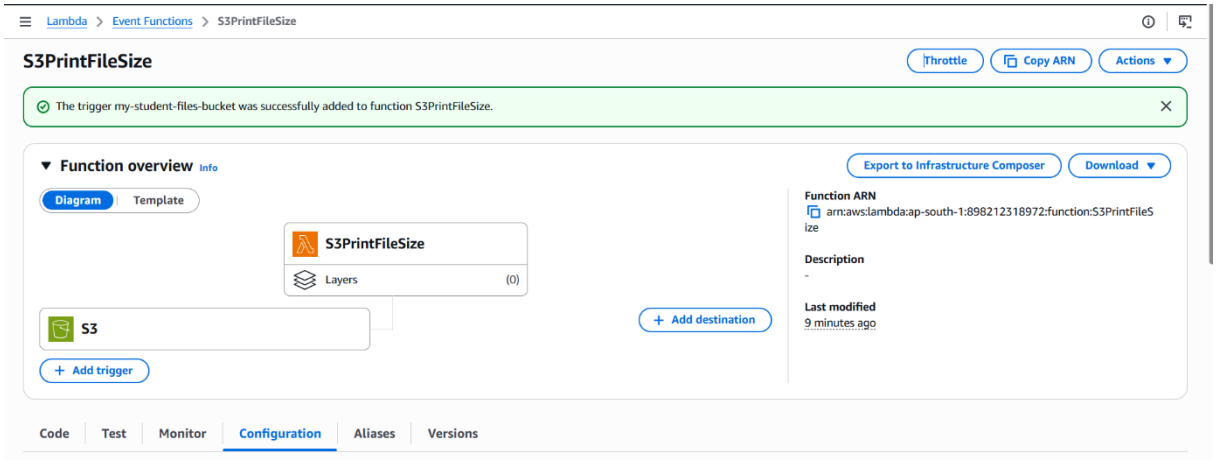


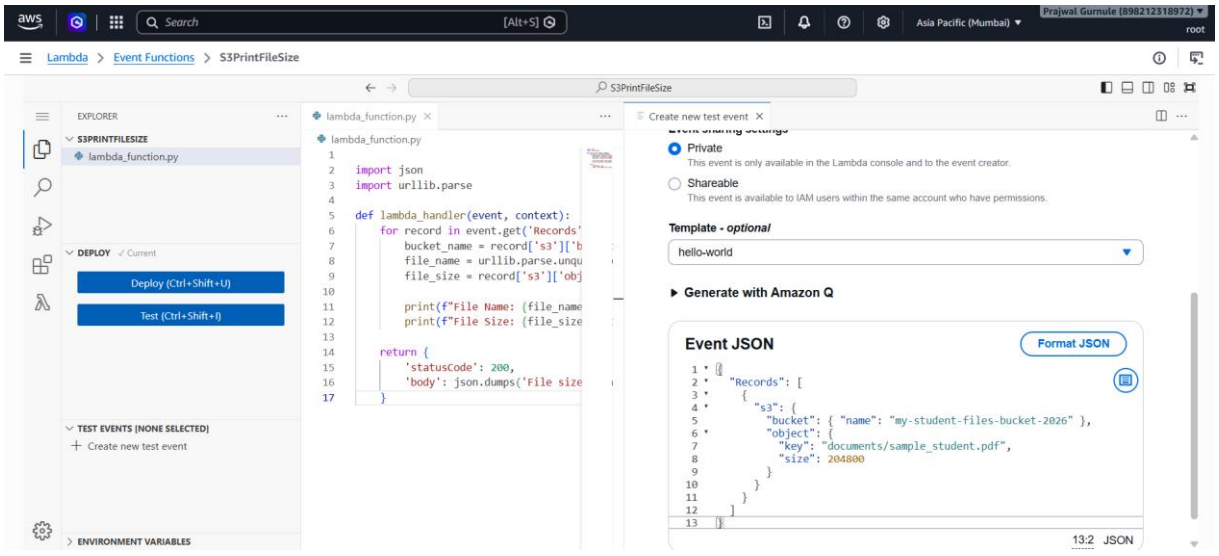
10. Create a Lambda function that triggers on S3 upload and prints the size of the uploaded file.

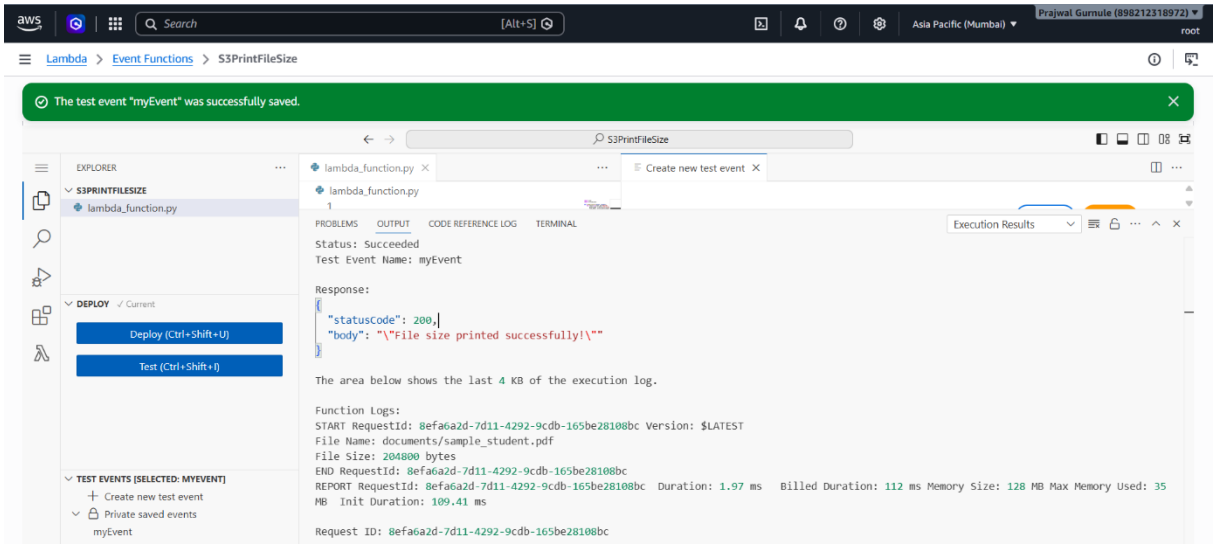


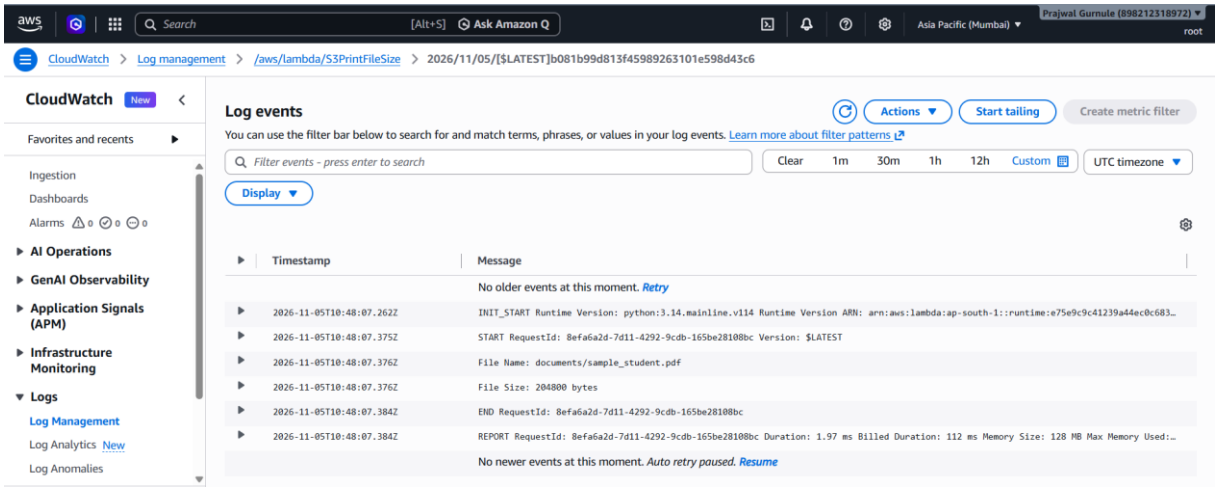



Search
[Alt+S]
Asia Pacific (Mumbai)
Prajwal Gurnule (898212318972)
root
Lambda > Add triggers
Add trigger
Trigger configuration Info
S3
aws asynchronous storage
Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.
s3/my-student-files-bucket
Bucket region: ap-south-1
Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.
All object create events
Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.
e.g. images/
Suffix - optional
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters. Any special characters must be URL encoded.


Lambda > Event Functions > S3PrintFileSize
S3PrintFileSize
Throttle
Copy ARN
Actions
The trigger my-student-files-bucket was successfully added to function S3PrintFileSize.
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
S3PrintFileSize
Layers (0)
S3
Add destination
Add trigger
Function ARN
arn:aws:lambda:ap-south-1:898212318972:function:S3PrintFileSize
Description
-
Last modified
9 minutes ago
Code
Test
Monitor
Configuration
Aliases
Versions


Search
[Alt+S]
Asia Pacific (Mumbai)
Prajwal Gurnule (898212318972)
root
Lambda > Event Functions > S3PrintFileSize
S3PrintFileSize
EXPLORER
S3PRINTFILESIZE
lambda_function.py
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [NONE SELECTED]
Create new test event
ENVIRONMENT VARIABLES
lambda_function.py
import json
import urllib.parse
def lambda_handler(event, context):
for record in event.get('Records'
bucket_name = record['s3']['b
file_name = urllib.parse.unqu
file_size = record['s3']['obj
print(f"File Name: {file_name
print(f"File Size: {file_size
return {
'statusCode': 200,
'body': json.dumps('File size
}
Create new test event
Private
This event is only available in the Lambda console and to the event creator.
Shareable
This event is available to IAM users within the same account who have permissions.
Template - optional
hello-world
Generate with Amazon Q
Event JSON
Format JSON
{
"Records": [
{
"s3": {
"bucket": { "name": "my-student-files-bucket-2026" },
"object": {
"key": "documents/sample_student.pdf",
"size": 204800
}
}
}
]
}
13:2 JSON


Search
[Alt+S]
Asia Pacific (Mumbai)
Prajwal Gurnule (898212318972)
root
Lambda > Event Functions > S3PrintFileSize
The test event "myEvent" was successfully saved.
S3PrintFileSize
EXPLORER
S3PRINTFILESIZE
lambda_function.py
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [SELECTED: MYEVENT]
Create new test event
Private saved events
myEvent
lambda_function.py
Create new test event
PROBLEMS
OUTPUT
CODE REFERENCE LOG
TERMINAL
Execution Results
Status: Succeeded
Test Event Name: myEvent
Response:
{
"statusCode": 200,
"body": "\"File size printed successfully!\""
}
The area below shows the last 4 KB of the execution log.
Function Logs:
START RequestId: 8efa6a2d-7d11-4292-9cdb-165be28108bc Version: $LATEST
File Name: documents/sample_student.pdf
File Size: 204800 bytes
END RequestId: 8efa6a2d-7d11-4292-9cdb-165be28108bc
REPORT RequestId: 8efa6a2d-7d11-4292-9cdb-165be28108bc Duration: 1.97 ms Billed Duration: 112 ms Memory Size: 128 MB Max Memory Used: 35 MB Init Duration: 109.41 ms
Request ID: 8efa6a2d-7d11-4292-9cdb-165be28108bc


Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Prajwal Gurnule (898212318972)
root
CloudWatch > Log management > /aws/lambda/S3PrintFileSize > 2026/11/05/[$LATEST]b081b99d813f45989263101e598d43c6
CloudWatch
New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
Log Analytics New
Log Anomalies
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear
1m
30m
1h
12h
Custom
UTC timezone
Display
Timestamp
Message
No older events at this moment. Retry
2026-11-05T10:48:07.262Z INIT_START Runtime Version: python:3.14.mainline.v114 Runtime Version ARN: arn:aws:lambda:ap-south-1::runtime:e75e9c9c41239a44ec0c683…
2026-11-05T10:48:07.375Z START RequestId: 8efa6a2d-7d11-4292-9cdb-165be28108bc Version: $LATEST
2026-11-05T10:48:07.376Z File Name: documents/sample_student.pdf
2026-11-05T10:48:07.376Z File Size: 204800 bytes
2026-11-05T10:48:07.384Z END RequestId: 8efa6a2d-7d11-4292-9cdb-165be28108bc
2026-11-05T10:48:07.384Z REPORT RequestId: 8efa6a2d-7d11-4292-9cdb-165be28108bc Duration: 1.97 ms Billed Duration: 112 ms Memory Size: 128 MB Max Memory Used:…
No newer events at this moment. Auto retry paused. Resume